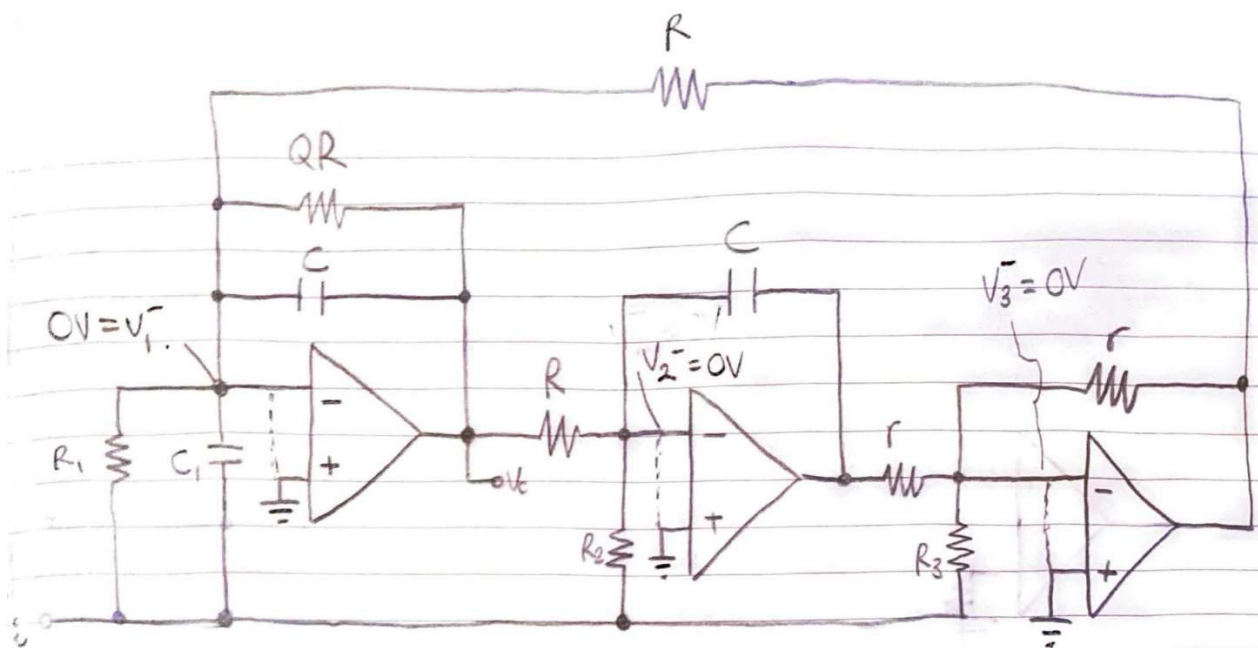




$$V_i: \frac{V_i - 0}{R_1} + \frac{V_i - 0}{1/sC_1} + \frac{V_i - 0}{R_2} + \frac{V_i - 0}{R_3} = 0$$

$$V_1^-: \frac{V_1^- - V_i}{R_1} + \frac{V_1^- - V_i}{1/sC_1} + \frac{V_1^- - V_0}{1/sC} + \frac{V_1^- - V_0}{QR} + \frac{V_1^- - V_{03}}{R} = 0$$

$$V_2^-: \frac{V_2^- - V_0}{R} + \frac{V_2^- - V_i}{R_2} + \frac{V_2^- - V_{02}}{1/sC} = 0$$

$$V_3^-: \frac{V_3^- - V_{02}}{r} + \frac{V_3^- - V_i}{R_3} + \frac{V_3^- - V_{03}}{r} = 0$$

$$V_1^-: \frac{V_i}{R_1} + V_i sC_1 + V_0 sC + \frac{V_0}{QR} + \frac{V_{03}}{R} = 0$$

$$V_2^-: \frac{V_0}{R} + \frac{V_i}{R_2} + V_{02} sC = 0$$

$$V_3^-: \frac{V_{02}}{r} + \frac{V_i}{R_3} + \frac{V_{03}}{r} = 0$$

$$Q_1) \quad \frac{V_{o2}}{R} + \frac{V_i}{R_2} + V_{o2} sC = 0$$

$$V_{o2} = -\frac{1}{sC} \left(\frac{V_o}{R} + \frac{V_i}{R_2} \right)$$

$$\frac{V_{o2}}{r} + \frac{V_i}{R_3} + \frac{V_{o3}}{r} = 0$$

$$V_{o3} = -r \left(\frac{V_{o2}}{r} + \frac{V_i}{R_3} \right) = -V_{o2} - \frac{r}{R_3} V_i$$

$$= \frac{1}{sC} \left(\frac{V_o}{R} + \frac{V_i}{R_2} \right) - \frac{r}{R_3} V_i$$

$$\text{Into } V_i^-: \frac{V_i}{R_1} + V_i sC_1 + V_o sC + \frac{V_i}{QR} + \frac{1}{R} \left[-\frac{1}{sC} \left(\frac{V_o}{R} + \frac{V_i}{R_2} \right) - \frac{r}{R_3} V_i \right]$$

Searched to simplify:

$$\frac{V_o}{V_i} = \frac{s^2 \frac{C_1}{C} + s \left(\frac{1}{RC} - \frac{r}{R_3 RC} \right) + \frac{1}{R_2 RC^2}}{(s^2 + s \frac{1}{RCQ} + \frac{1}{R^2 C^2})} \rightarrow = "d"$$

$$a) \quad C_1 \rightarrow 0, \quad \frac{1}{R_1 C} = \frac{r}{R_3 C} \Rightarrow R_1 = \frac{R_3 R}{r} \rightarrow \frac{R_3 RC^2}{d}$$

b) Cant, constant not configurable

c) Same as above

$$d) \quad R_1 = R_3 R / r \Rightarrow \frac{s^2 \frac{C_1}{C} + \frac{1}{R_2 RC^2}}{d}$$

Q₂) for gain of 1:

$$R = 10k \rightarrow R_g = \underline{10k} \approx 10k\Omega$$

$$R_d = 10k(20) = \underline{200k\Omega}$$

$$R_g = R_d = 200k$$

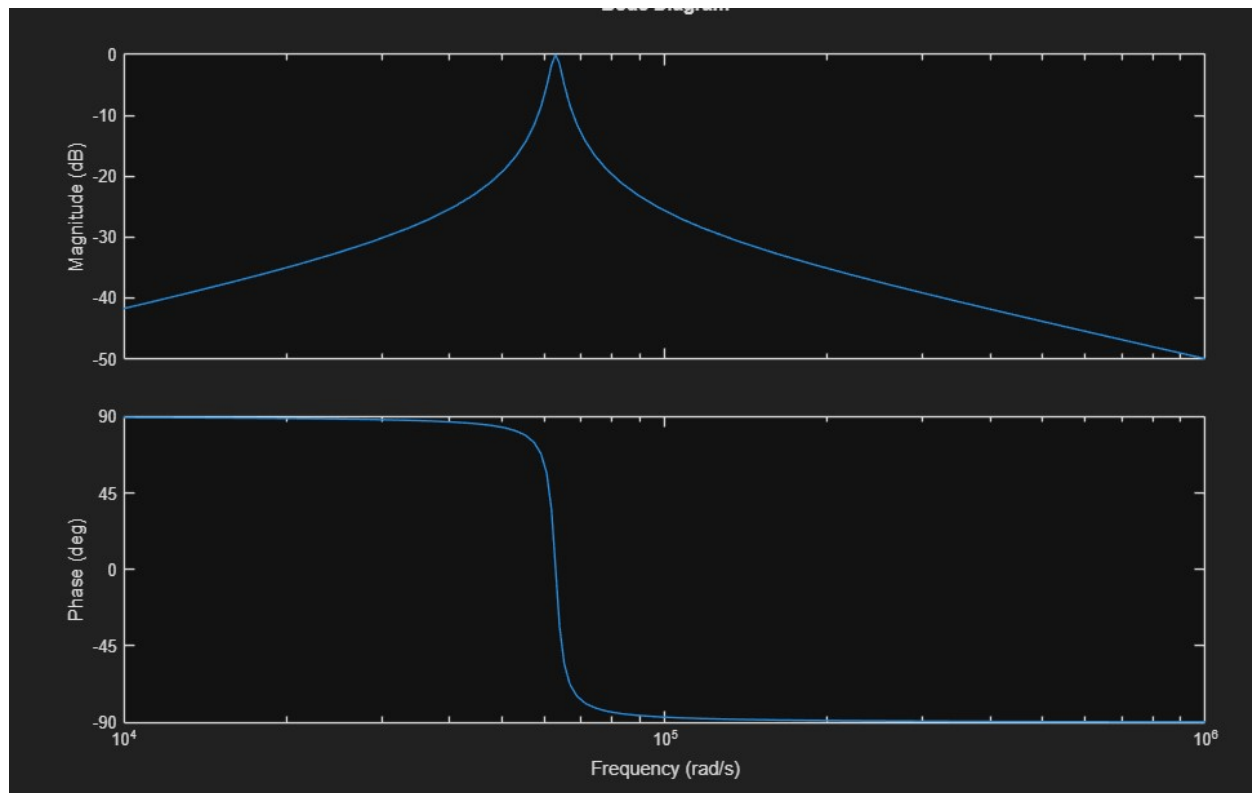
$$C = \frac{1}{2\pi f_c R} = \frac{1}{2\pi(10^8)} = 1.59nF$$

$$H(s) = \frac{s \frac{1}{R_g C}}{s^2 + s \frac{1}{R_d C} + \frac{1}{R_d^2 C^2}}$$

$$Q_3) -\frac{V_o}{R_F} - \frac{V_{HP}}{R_H} = \frac{V_{BP}}{R_B} - \frac{V_{LP}}{R_L} = 0$$

$$V_o = -R_F \left(\frac{V_{HP}}{R_H} + \frac{V_{BP}}{R_B} + \frac{V_{LP}}{R_L} \right)$$

$$\frac{V_o}{V_i} = \left(\frac{R_F}{R_L} \right) \left(\frac{R_F}{R_H} \right) \left[\frac{s^2 - s\omega_0 \frac{R_H}{R_B} + \omega_0^2 \frac{R_H}{R_L}}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2} \right]$$



```
clear all; close all; clc;
```

```
R=10000;
```

```
Rd=200000;
```

```
Rg=200000;
```

```
C=1.59*10^-9;
```

```
num = [1/(Rg*C) 0];
```

```
den = [1 1/(Rd*C) 1/(R^2 * C^2)];
```

```
sys = tf(num, den);
```

```
bode(sys);
```